



AGRICULTURE — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Permaculture vs Conventional Chemical Farming
2. Zero Tillage — Advantages
3. Fertigation — Benefits & Limitation
4. Eco-Friendly Agriculture Practices
5. M.S. Randhawa — Agriculture History Author
6. Agro-Climatic vs Agro-Ecological Regions (15 vs 20)
7. Upper Gangetic Agro-Climatic Region
8. P. Sengupta & Sdasyuk — 60 Micro Agricultural Regions
9. Dry Zone — Second Green Revolution Potential
10. Seed Replacement Rate — Implementation Constraints
11. First Agricultural University — Pantnagar (1960)
12. Safe Grain Storage — 14% Moisture Limit
13. Land Use Classification (Net Sown/Forest/Other)
14. Double Cropping vs Intercropping vs Mixed Farming
15. Parallel Cropping — Wheat+Mustard Example
16. Sikkim — Lowest Agricultural Land %
17. Characteristics of Indian Agriculture (Small Holdings)
18. Low Productivity Causes (Overcrowding, Small Holdings)
19. Green Revolution — Environmental Cost (Assertion-Reason)
20. Cooperative Farming — Not a Low-Productivity Cause
21. Largest Average Landholding — Rajasthan
22. Agriculture as Livelihood (Not Just Profession)
23. Pulses — Rainfed Area Statistics (90%)
24. Fertilizer Consumption Leaders (UP & Andhra Pradesh)
25. Green Manure Crops — Nitrogen Content Ranking
26. Balanced Fertilizers — Purpose
27. High Agricultural Productivity Areas — South India
28. Groundwater Resources — UP Richest
29. Contract Farming — Punjab Pioneer
30. Green Agriculture — Integrated Management Concept
31. Globalization's Impact on Indian Agriculture
32. Seed Village Concept
33. Agmark — Quality Certification (1937 Act)
34. Landholding Size Classification (Marginal to Big)
35. Conservation Agriculture — Components
36. Pathways to Increase Agricultural Productivity
37. Agriculture Census 2015-16 — Key Statistics

1. Permaculture vs Conventional Chemical Farming

- Permaculture farming **monoculture ko discourage karti hai, mulching bahut important hai, aur semi-arid regions mein bhi effective hai** — chemical farming mein monoculture dominant hai. 'Permaculture semi-arid regions mein achhi tarah possible nahi hai' **GALAT statement hai.** [Q1]

2. Zero Tillage — Advantages

- Zero tillage ke sab 3 advantages sahi hain: **residue jalaye bina wheat sowing, direct paddy seeding (nursery ke bina), carbon sequestration se greenhouse effect kam.** [Q2]

3. Fertigation — Benefits & Limitation

- Fertigation (drip system se paani mein fertilizer milana) ke fayde: **alkalinity control, nutrients ki increased availability, chemical nutrients ki leaching mein kami — 'Rock Phosphate ka efficient application'** GALAT hai (super phosphorus se phosphate precipitation hoti hai, isliye avoid karna chahiye). [Q3]

4. Eco-Friendly Agriculture Practices

- Eco-friendly practices sab 4 hain: **crop diversification, legume intensification (nitrogen-fixing), tensiometer use (soil moisture estimate), vertical farming.** [Q4]

5. M.S. Randhawa — Agriculture History Author

- 'The History of Indian Agriculture' **M.S. Randhawa (Mohinder Singh Randhawa)** ne likhi hai — Green Revolution mein bhi bada role tha. [Q5]

6. Agro-Climatic vs Agro-Ecological Regions (15 vs 20)

- India ko **resource development ke liye 15 agro-climatic regions** mein baanta gaya hai — **National Bureau of Soil Survey ke mutabik 20 agro-ecological zones hain (17/18/19 GALAT hain).** Western Himalaya cold arid ecoregion ka coverage area, Western Himalaya region se zyada hai. [Q6, Q8, Q9]

7. Upper Gangetic Agro-Climatic Region

- Upper Gangetic region mein **Central-Western UP** شامل hai — **irrigation intensity ~131%, cropping intensity ~142%** — 4 sub-zones (central-western plain, SW semi-arid, western plain, central plain) hain. [Q7]

8. P. Sengupta & Sdasyuk — 60 Micro Agricultural Regions

- 1968 mein Registrar General of Census ne **Dr. P. Sengupta aur Dr. Galina Sdasyuk** ke study ke basis par **India ko 60 micro agricultural regions** mein baanta. [Q10]

9. Dry Zone — Second Green Revolution Potential

- India ke dry zone mein **agrarian economy hai (Assertion sahi)** — **dusri Green Revolution ki bhi bahut potential hai (Reason sahi),** lekin Reason, Assertion ki sahi explanation NAHI hai. [Q11]

10. Seed Replacement Rate — Implementation Constraints

- Seed Replacement Rate ke wider implementation mein constraint: **low-value/high-volume crops ke liye quality seeds ki demand-supply gap — 'National Seeds Policy nahi hai' aur 'private sector companies ki participation nahi hai'** dono GALAT statements hain (2002 mein hi National Seed Policy ban chuki thi). [Q12]

11. First Agricultural University — Pantnagar (1960)

- India ka pehla Agricultural University **G.B. Pant University of Agriculture and Technology, Pantnagar (Uttarakhand)** hai — **1960** mein **Jawaharlal Nehru** ne 'UP Agricultural University' ke naam se **inaugurate kiya tha**, baad mein UP ke pehle CM ke naam par rename hua. [Q13, Q14, Q15]

12. Safe Grain Storage — 14% Moisture Limit

- FAO standards ke mutabik safe storage ke liye **harvest ke time grain moisture content 14% se zyada nahi honi chahiye**. [Q16]

13. Land Use Classification (Net Sown/Forest/Other)

- India ka land-use classification (2021-22): **Net Area Sown 46.01%, Forest Area 23.49%, Other Areas 30.5%**. [Q17]

14. Double Cropping vs Intercropping vs Mixed Farming

- Double cropping — **do crops alag-alag time par (same year mein) same land par** — Intercropping — **do ya zyada crops SAATH mein same field mein**. Mixed farming — **animals aur crops dono ek saath rear karna (equal emphasis)**. [Q18, Q19, Q20]

15. Parallel Cropping — Wheat+Mustard Example

- Parallel cropping (dono crops nutrients ke liye compete nahi karte, ek dusre par asar nahi daalte) ka example **Wheat + Mustard** hai. [Q21]

16. Sikkim — Lowest Agricultural Land %

- Given states (Punjab, Haryana, UP, Sikkim) mein **Sikkim mein sabse kam agricultural land (10% se kam)** hai — hilly, forest-covered state hai. [Q22]

17. Characteristics of Indian Agriculture (Small Holdings)

- Indian agriculture ki characteristics: **nature par over-dependence, low productivity, crop diversity** — 'large farms ka dominance' **GALAT** hai (India mein small/marginal holdings ka dominance hai). [Q23]

18. Low Productivity Causes (Overcrowding, Small Holdings)

- Indian agriculture mein low productivity ke reasons: **overcrowding, small land holding, traditional practices** — sab teeno sahi hain. [Q24]

19. Green Revolution — Environmental Cost (Assertion-Reason)

- Green Revolution technology ne traditional agriculture ko modern-scientific banaya (Assertion sahi) — **lekin isme social/environmental costs shamil NAHI the** — **ye statement GALAT hai (environmental cost long-term sustainability ke liye threat maana jaata hai)** — dono statements sahi hain, par R, A ki sahi explanation nahi hai. [Q25]

20. Cooperative Farming — Not a Low-Productivity Cause

- Low productivity ke reasons: **population pressure, disguised unemployment, small land holding** — 'Cooperative farming' iska cause NAHI hai (ye actually development ka ek tareeka hai). [Q26]

21. Largest Average Landholding — Rajasthan

- India mein **operational holdings ka average size Rajasthan mein sabse bada** hai (2015-16) — Maharashtra, UP se zyada. [Q27]

22. Agriculture as Livelihood (Not Just Profession)

- India mein agriculture ko '**means of livelihood**' (**aajeevika ka saadhan**) maana jaata hai — maximum manpower isme lagi hai, par GDP mein sirf chhota hissa deti hai. [Q28]

23. Pulses — Rainfed Area Statistics (90%)

- Sahi statement: **India mein pulses ka ~90% area rainfed hai** — '**pulses ka share 2 dashak mein double hua**' GALAT hai (asal mein ghata hai), '**rice cultivation ka 15% share**' GALAT hai (~29% hai), '**rice 34% gross cropped area**' GALAT hai (~23.7% hai). [Q29]

24. Fertilizer Consumption Leaders (UP & Andhra Pradesh)

- Question ke waqt (2013-14) fertilizer ke **2 sabse bade consumers** — **Uttar Pradesh aur Andhra Pradesh** the — abhi (2021-22) UP, Maharashtra, MP top-3 hain. [Q30]

25. Green Manure Crops — Nitrogen Content Ranking

- Nitrogen content ranking (green manure crops): **Cow pea (0.49%) > Sunhemp (0.43%) > Dhaincha (0.42%) > Guar (0.34%)**. Newly-improved arid land ke liye **Lobia (cow pea)** sabse suitable green manure crop hai. [Q31, Q32]

26. Balanced Fertilizers — Purpose

- Balanced fertilizers **production badhaane, quality improve karne, aur soil productivity maintain karne** — teeno purpose ke liye use hote hain. [Q33]

27. High Agricultural Productivity Areas — South India

- South India mein high agricultural productivity **Tamil Nadu coast** mein hai (Andhra Pradesh coast, Surat-Gujarat, Satara-Kolhapur-Maharashtra bhi high productivity areas hain). [Q34]

28. Groundwater Resources — UP Richest

- Replenishable groundwater resources mein **Uttar Pradesh sabse rich state** hai (2024 data: UP 72.84 bcm, MP 35.90, AP 27.80, WB 25.89). [Q35]

29. Contract Farming — Punjab Pioneer

- India mein contract farming introduce karne ka **pioneer Punjab** hai. [Q36]

30. Green Agriculture — Integrated Management Concept

- 'Green Agriculture' involves: **integrated pest management, integrated nutrient supply, integrated natural resource management** — sirf organic farming ya sirf horticulture NAHI. [Q37]

31. Globalization's Impact on Indian Agriculture

- 1991 mein Narasimha Rao Government ne Globalization introduce ki — impacts: **cash crops par emphasis, income inequality mein growth, subsidies mein kami** — 'Climate change' iska direct impact NAHI tha (galat option). [Q38]

32. Seed Village Concept

- Seed Village Concept ka objective: **farmers ko quality-seed-production training dena, taaki appropriate time aur affordable cost par doosre farmers ko bhi quality seeds mil sakein**. [Q39]

33. Agmark — Quality Certification (1937 Act)

- Agmark **quality certification mark hai** (Agricultural Produce Grading and Marking Act, 1937 ke tahat) — 1938 se India mein lagu hai. [Q40]

34. Landholding Size Classification (Marginal to Big)

- Landholding size classification (2015-16): **Marginal — <1 hectare, Small — 1-2 hectares, Semi-medium — 2-4 hectares, Medium — 4-10 hectares, Big — >10 hectares**. [Q41]

35. Conservation Agriculture — Components

- Conservation Agriculture ke sahi components: **minimum tillage adopt karna, crop residues se soil surface cover karna, spatial/temporal crop rotation apnaana** — 'monoculture avoid karna' aur 'plantation crops avoid karna' iske components NAHI hain. [Q42]

36. Pathways to Increase Agricultural Productivity

- Agriculture productivity badhaane ke sab tareeke sahi hain: **efficient irrigation, quality seeds, pesticides use, fertilizers use** — sab combined. [Q43]

37. Agriculture Census 2015-16 — Key Statistics

- 2015-16 Agriculture Census ke sahi facts: **small+marginal farmers 86.2% (kul farmers ka) hain lekin sirf 47.3% land hold karte hain, Rajasthan total-land ke hisaab se 1st hai, Maharashtra total-farmers ke hisaab se 3rd hai** — sab teeno statements sahi hain. [Q44]